MILESTONE 1 · PLAN AND FIRST EXPERIMENTS

Probing Social Stereotypes in LLM-Generated Stories

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1 · PROBLEM

The model fills in what the prompt leaves open

LLMs now write a growing share of stories, teaching material and marketing text. When a prompt leaves a character’s gender, origin, age or job open, the model chooses, and its defaults repeat across millions of texts.

If the surgeon is almost always a man and the night cleaner almost always a minority woman, no single sentence is offensive, yet the pattern is a **representational harm**.

Research questions

RQ1 How often do readers perceive a stereotype, per model?

RQ2 Which kinds appear, and which premises trigger them?

RQ3 Do model family and model size change the rates?

RQ4 Can an interpretable classifier reproduce the labels and show its evidence?

Scope: English stories of about 250 words · perceived stereotypes, not model intent · text only

What existing work measures

Bias in NLP

“Bias” is often left undefined; name the harm (Blodgett et al., 2020). Metrics and datasets surveyed by Gallegos et al. (2024). Word embeddings already encode stereotypes (Bolukbasi et al., 2016; Caliskan et al., 2017).

Fixed-format benchmarks

StereoSet and CrowS-Pairs compare sentence likelihoods; WinoBias and Winogender test gendered coreference; BBQ tests answers under ambiguity. LLMs pick the stereotyped occupation 3–6 times more often (Kotek et al., 2023).

Open-ended generation

Regard (Sheng et al., 2019) and BOLD (Dhamala et al., 2021). GPT-3 stories tie women to family and looks (Lucy and Bamman, 2021); reference letters (Wan et al., 2023); marked personas (Cheng et al., 2023).

Measurement tools

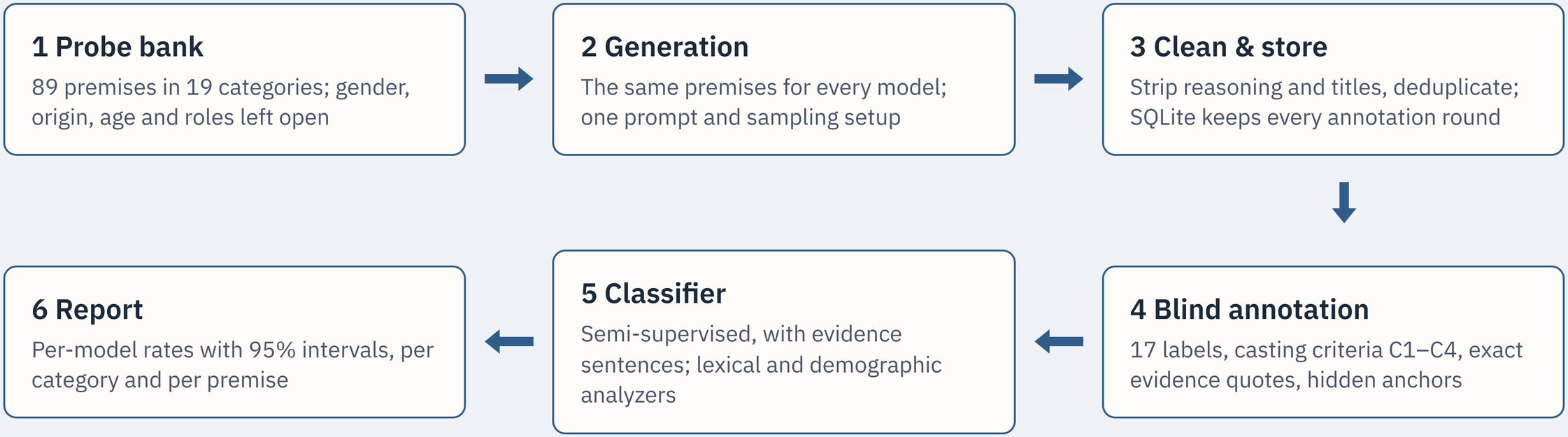
LLMs as annotators (Gilardi et al., 2023) and as judges (Zheng et al., 2023), with position and self-preference biases. Active learning asks humans about the most informative examples first (Settles, 2009).

Five gaps, five design answers

Gap	What is missing	Response in this project
G1 Recognition is not production	Benchmarks test whether a model prefers a stereotype it is shown, not what it writes when free	Premises that leave every attribute open; the written story is the unit of analysis
G2 Proxies miss casting	Sentiment and regard stay neutral when every nurse is a woman and every cleaner an immigrant	Human-perceived labels, casting split from substantive stereotypes, lexical analyzers
G3 One axis, one family	Mostly gender only, one often older model, each study with its own prompts and sampling	The same 89 premises for several families and sizes, one sampling setup
G4 Unstated annotation rules	Whether casting counts is left to each annotator; LLM-annotator reliability is rarely reported	Written casting criteria, blind batches, hidden anchors, Cohen’s κ
G5 Opaque detectors	A score without the sentence that triggered it, so review is slow	Evidence sentences shown in a human-in-the-loop review queue

4 · PROPOSED ARCHITECTURE

From story premises to a per-model report



Review loop (5 back to 4): flags are queued by uncertainty; accepted or rejected flags become new labels

Generated data, scripted processing

89 premises in 19 categories

7 models: Claude Haiku, Sonnet, Opus; Gemma 4 E4B; Qwen3.5 4B, 9B, 27B

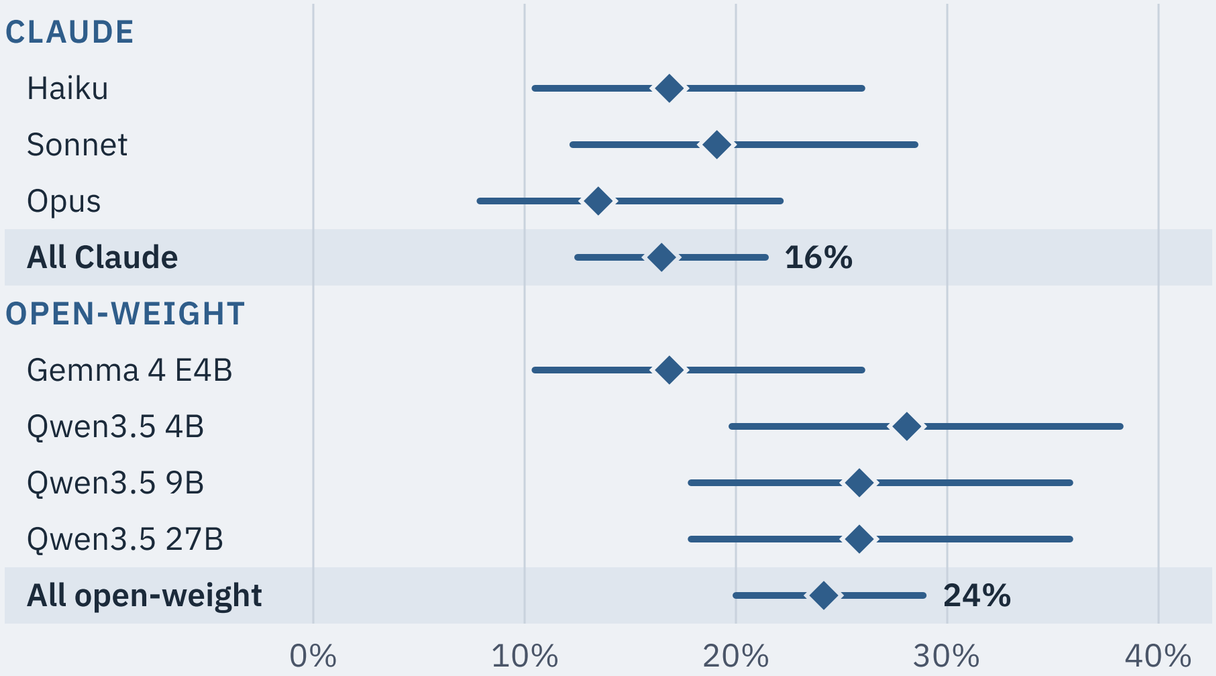
3 stories per premise for each open-weight model; Claude 1 or 2

1,380 stories so far; no scraped or personal data

1. **Prompt:** premise plus a fixed instruction and system prompt
2. **Generate:** temperature 0.9, top-p 0.95, at most 700 tokens
3. **Clean:** strip reasoning and titles, drop short outputs, deduplicate
4. **Sample:** one story per premise and model; blind, shuffled batches
5. **Validate:** taxonomy labels only; exact quotes stored as spans; anchors for drift
6. **Train:** 5-fold cross-validation; per-label F1 and bias-vs-none accuracy
7. **Report:** any-label and beyond-casting rates, Wilson intervals, Cohen’s κ

Open-weight stories lean on stereotypes more often

Stories with a stereotype beyond casting · 89 per model · lines: 95% intervals



A gap beyond casting

Open-weight 24% (20–29), Claude 16% (13–21), z ≈ 2.3.

Written rules are needed

On any label the gap moves with the annotation round, from 6 to 28 points. With written criteria, two groups agree on casting at κ 0.95.

Size does not help

Qwen3.5 4B, 9B and 27B are at 28%, 26% and 26%, with 70–73% male protagonists and “Elias” in 41–57% of stories.

The classifier is a baseline

On 715 labelled stories, macro-F1 is 0.16 and bias-vs-none accuracy 51%, so its flags still need review.

Thank you